

RAHUL MITRA

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EDUCATION

National University of Singapore

Singapore

Bachelor of Engineering, Industrial Systems Engineering

Aug 2023 – Jul 2027

- Pursuing a Second Major in Computer Science and a Minor in Mathematics.
- Awarded Certificate of Outstanding Performance in 3D Computer Vision (CS4277) — top student in a class of 24, NUS School of Computing.
- Relevant coursework: Software Engineering, Artificial Intelligence, 3D Computer Vision, Data Structures and Algorithms, Simulation, Statistics, Operations Research.

EXPERIENCE

STMicroelectronics

Singapore

Operations Research (NUS System Design Project), Contract

Aug 2026 – Present

- Build an AI-driven put-away recommendation system for STMicroelectronics' Singapore warehouse, combining picking history, demand forecasts, and live capacity to replace experience-based slotting decisions.

Amazon

Singapore

Robotics Vision Engineer (Blended AI+X)

Jul 2026 – Present

- Build vision systems for camera ISP enhancement, super-resolution, and image restoration under motion and low light.

Abbott Laboratories

Singapore

Operational AI Systems & Data Engineer (Contract)

Jun 2026 – Present

- Build AI-enabled systems and data workflows for process improvement, manufacturing analytics, operational decision support, and supply-chain automation.
- Strengthen monitoring and process structure across healthcare supply chains with AI, helping ensure reliable product availability for vulnerable patient groups.

Abbott Laboratories

Singapore

Data Analytics and Project/Product Management Intern

Jan 2026 – Jun 2026

- Productionized a team-built APC simulator for live manufacturing and engineer training, packaging it in Docker and deploying it to Azure App Service.

People's Association

Singapore

Platform & Solutions Engineer (Sparks Citizen Developer)

Sep 2025 – Sep 2026

- Built an end-to-end digital garden-plot balloting platform spanning admin and public frontends, backend, GIS mapping, and deployment, now production-ready for national rollout across Singapore.
- Integrated Singpass Login (OIDC) and Myinfo for passwordless authentication and consent-based retrieval of verified identity and registered address, automating resident eligibility checks.
- Led deployment planning, multi-repo coordination, and stakeholder management across product, engineering, and operations; awarded the S\$20,000 Sparks Community Innovation Fund.
- Provisioned and deployed the platform's AWS infrastructure with Terraform — Fargate services, Route 53 DNS, Redis caching, and Kafka messaging — applying cloud security controls across the environment.

PROJECTS

Volt Pulse SG — SMU Hack For Cities 2026 (Top 8 Finalist of 50+ teams)

Jan 2026

- Built an agentic AI system for tracking HDB household energy costs using SEALION embeddings, Supabase vector search, RRF routing, and scheduler-agent planning.

AgeWellLah.AI — NUS HealthHack 2025

2025

- Built an elderly-care triage chatbot using OpenAI GPT-4, InterSystems IRIS Vector Search, and Flask, with OAuth2-secured patient profiles and PDPA-aware design.

LEADERSHIP AND ACTIVITIES

NTUC Health Active Ageing Centre

Singapore

Elderly Care Volunteer

Sep 2025 – Present

- Support weekly senior-engagement activities, deepening empathy for healthcare, accessibility, and human-centered community product design.

SKILLS AND CERTIFICATIONS

Technical: Solution architecture; cloud (AWS Fargate, Terraform, Azure, Docker, CI/CD); full-stack (React, TypeScript, FastAPI, Flask); Singpass Login & Myinfo (OIDC); RAG & AI agents; GIS mapping; accessibility; rapid prototyping

Certifications: NVIDIA Disaster Risk Monitoring with Satellite Imagery; Deep Reinforcement Learning Hands-On (Packt)